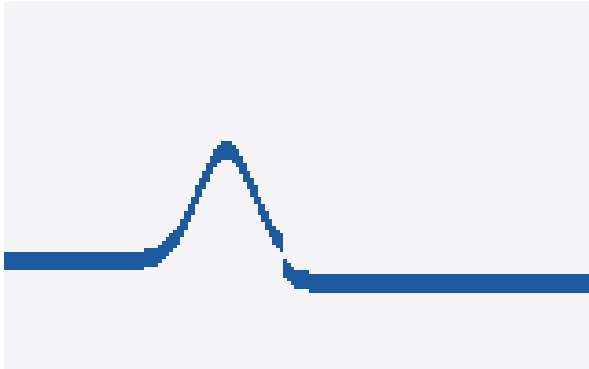


## Lecture 7 — Action Potentials

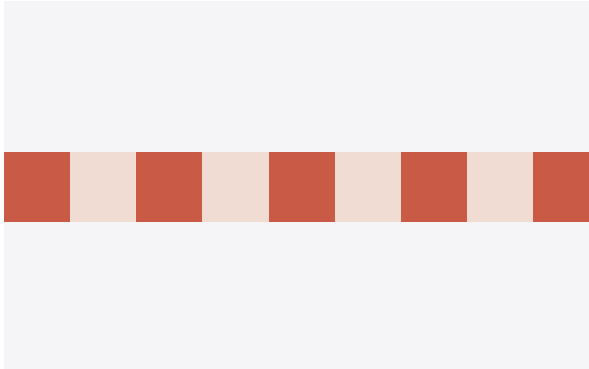
The resting membrane potential of a typical neurone sits at approximately  $-70$  mV. It is established by the  $\text{Na}^+/\text{K}^+$  ATPase, which exports three sodium ions for every two potassium ions imported, together with the differential permeability of the axolemma to potassium at rest.



*Figure 1. Membrane potential trace during depolarisation and repolarisation.*

## Saltatory conduction

Voltage-gated sodium channels open once the axon hillock reaches threshold, producing rapid depolarisation. Myelination increases conduction velocity because the action potential regenerates only at the nodes of Ranvier rather than propagating continuously. This is examinable: contrast conduction velocity in myelinated and unmyelinated fibres.



*Figure 2. Saltatory conduction along a myelinated axon.*